

|                                                        |                                                       |
|--------------------------------------------------------|-------------------------------------------------------|
| Program : <b>Diploma in Robotic Process Automation</b> |                                                       |
| Course Code : <b>5301</b>                              | Course Title: <b>Introduction to Machine Learning</b> |
| Semester: <b>5</b>                                     | Credits: <b>4</b>                                     |
| Course Category: <b>Program Core</b>                   |                                                       |
| Periods per week: <b>4 (L: 4 T: 0 P: 0)</b>            | Periods per semester: <b>60</b>                       |

#### Course Objectives:

- To introduce the prominent methods for machine learning
- To study the basics of supervised and unsupervised learning
- To study the basics of connectionist and other architectures

#### Course Prerequisites:

| Topic                                        | Course code | Course name       | Semester |
|----------------------------------------------|-------------|-------------------|----------|
| Vector Calculus, Linear Algebra, Probability | 1002 & 2002 | Mathematics I &II | 1 & 2    |

#### Course Outcomes:

On completion of the course student will be able to:

| CO <sub>n</sub> | Description                                                                                     | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | Differentiate various learning approaches, and to interpret the concepts of supervised learning | 12               | Understanding   |
| <b>CO2</b>      | Compare the different dimensionality reduction techniques                                       | 14               | Applying        |

|            |                                                  |    |          |
|------------|--------------------------------------------------|----|----------|
| <b>CO3</b> | Demonstrate the basic concepts of classification | 15 | Applying |
| <b>CO4</b> | Study neural network and markov processes        | 17 | Applying |
|            | Series Test                                      | 2  |          |

### CO – PO Mapping:

| <b>Course Outcomes</b> | <b>PO 1</b> | <b>PO 2</b> | <b>PO 3</b> | <b>PO 4</b> | <b>PO 5</b> | <b>PO 6</b> | <b>PO 7</b> |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| CO1                    | 3           |             |             | 2           | 3           | 3           | 2           |
| CO2                    | 3           |             |             | 2           | 3           | 3           | 3           |
| CO3                    | 3           |             |             | 2           | 3           | 3           | 2           |
| CO4                    | 3           |             |             | 3           | 3           | 3           | 3           |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline

| <b>Module Outcomes</b> | <b>Description</b>                                                                                     | <b>Duration (Hours)</b> | <b>Cognitive Level</b> |
|------------------------|--------------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>             | <b>Differentiate various learning approaches, and to interpret the concepts of supervised learning</b> |                         |                        |
| M1.01                  | Introduction to Machine Learning, Examples of Machine                                                  | 2                       | Understanding          |
| M1.02                  | Learning applications - Learning associations                                                          | 2                       | Understanding          |
| M1.03                  | Classification, Regression                                                                             | 2                       | Understanding          |
| M1.04                  | Unsupervised Learning                                                                                  | 2                       | Understanding          |
| M1.05                  | Reinforcement Learning. Supervised learning-                                                           | 2                       | Understanding          |
| M1.06                  | - Input representation, Hypothesis class                                                               | 2                       | Understanding          |

Introduction to Machine Learning, Examples of Machine Learning applications - Learning associations, Classification, Regression, Unsupervised Learning, Reinforcement Learning. Supervised learning- Input representation, Hypothesis class, Version space, Vapnik-Chervonenkis (VC) Dimension

|            |                                                                  |   |               |
|------------|------------------------------------------------------------------|---|---------------|
| <b>CO2</b> | <b>Compare the different dimensionality reduction techniques</b> |   |               |
| M2.01      | Probably Approximately Learning (PAC)                            | 3 | Understanding |
| M2.02      | Noise, Learning Multiple classes,                                | 2 | Understanding |
| M2.03      | Model Selection and Generalization                               | 3 | Applying      |
| M2.04      | Dimensionality reduction- Subset selection                       | 3 | Applying      |
| M2.05      | Principle Component Analysis                                     | 3 | Applying      |
|            | Series Test – I                                                  | 1 |               |

**Contents:**

Probably Approximately Learning (PAC), Noise, Learning Multiple classes, Model Selection and Generalization, Dimensionality reduction- Subset selection, Principle Component Analysis

|            |                                                            |   |               |
|------------|------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>To Demonstrate the basic concepts of classification</b> |   |               |
| M3.01      | Classification- Cross validation and re-sampling methods   | 2 | Understanding |
| M3.02      | K- fold cross validation                                   | 2 | Understanding |
| M3.03      | Boot strapping                                             | 2 | Applying      |
| M3.04      | Measuring classifier performance- Precision, recall,       | 2 | Applying      |
| M3.05      | ROC curves.                                                | 2 | Understanding |
| M3.06      | Bayes Theorem, Bayesian classifier                         | 2 | Understanding |
| M3.07      | Maximum Likelihood estimation                              | 3 | Understanding |

**Contents:**

Classification- Cross validation and re-sampling methods- K- fold cross validation, Boot strapping, Measuring classifier performance- Precision, recall, ROC curves. Bayes Theorem, Bayesian classifier, Maximum Likelihood estimation Regression

|                                                                                                                                                                                                                                                                              |                                                       |   |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---|---------------|
| <b>CO4:</b>                                                                                                                                                                                                                                                                  | <b>Study neural network and markov processes</b>      |   |               |
| M4.01                                                                                                                                                                                                                                                                        | Neural Networks- The Perceptron, Activation Functions | 2 | Understanding |
| M4.02                                                                                                                                                                                                                                                                        | Support Vector Machine                                | 3 | Understanding |
| M4.03                                                                                                                                                                                                                                                                        | Discrete Markov Processes                             | 2 | Understanding |
| M4.04                                                                                                                                                                                                                                                                        | Hidden Markov models                                  | 2 | Understanding |
| M4.05                                                                                                                                                                                                                                                                        | Three basic problems of HMMs- Evaluation problem      | 2 | Understanding |
| M4.06                                                                                                                                                                                                                                                                        | Unsupervised Learning - Clustering Methods            | 3 | Understanding |
| M4.07                                                                                                                                                                                                                                                                        | Expectation-Maximization Algorithm,                   | 3 | Understanding |
|                                                                                                                                                                                                                                                                              | Series Test – II                                      | 1 |               |
| Neural Networks- The Perceptron, Activation Functions. Support Vector Machine . Discrete Markov Processes, Hidden Markov models, Three basic problems of HMMs- Evaluation problem, Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm. |                                                       |   |               |

### References:

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
2. Ethem Alpaydm, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004.
3. Margaret H. Dunham. Data Mining: introductory and Advanced Topics, Pearson, 2006
4. Mitchell. T, Machine Learning, McGraw Hill.
5. Ryszard S. Michalski, Jaime G. Carbonell, and Tom M. Mitchell, Machine Learning : An Artificial Intelligence Approach, Tioga Publishing Company.nce